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Data and Analytics in Accounting

AN INTEGRATED APPROACH

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An Integrated Approach

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Dedication:

This book would not be possible without the support of my family and inspiration from my students. To my husband, Dave, thank you for your patience and unwavering support. To my children Greg, and Jenny, I could not have done this without your words of encouragement and willingness to help when needed. To my siblings, nieces, nephews, great nieces, and great nephew, thank you for forgiving missed family events, birthdays, and special occasions. I love you all! —Ann Dzurani

To my parents for their strong commitment to my education and for so much more. To my wife and daughter for their patience and unconditional support of everything I do. To all my wonderful students, past and present, who inspire me with their curiosity and enthusiasm. —Guido Geerts

To my parents, Peter and Maria Lenk, my children, Nick, Tara, Andy, Andrea, and Elizabeth, and my three grandsons, and to my students and mentors, for their excellent examples of spirituality, love, integrity, family, vision, perseverance, generosity, and work ethic. I love you all and could not have done this without your support and help! —Margarita Lenk

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About the Authors



© Nick Porcuro

ANN C. DZURANI is the KPMG Endowed Professor of Accountancy at Northern Illinois University. She earned her BS from Fairleigh Dickinson University, her MBA from New York University, and her PhD from the University of South Florida. Ann is a CPA (NJ) with 15 years of experience in both public and corporate accounting.

Ann conducts behavioral research in managerial accounting decision-making and the ways in which accounting information systems interact with those decisions. Her publications include *Issues in Accounting Education*, *Journal of Information Systems*, *Journal of Business Ethics*, *Strategic Finance*, *Management Accounting Quarterly*, *IMA Statement on Management Accounting*, *Journal of Corporate Accounting and Finance* and the *Journal of Accounting Education*.

Ann received the 2018 American Accounting Association's Innovation in Accounting Education Award for her work in data analytics curriculum development. Ann's work in data analytics has resulted in invitations to present on Data Analytics and Accounting curriculum at both academic and professional conferences. Her presentations have reached over 3,000 people and her materials have been shared with more than 75 universities.

Ann served on the AICPA CPA Evolution Taskforce and currently serves as a BEC subcommittee member and a member of the CPA exam Content Committee.



© Guido Geerts

GUIDO L. GEERTS is a professor of accounting and EY Faculty Scholar at the Lerner College of Business, University of Delaware, where he teaches accounting information systems and data analytics. He received a PhD in accounting information systems from the Free University of Brussels, Belgium in 1993.

Guido's research has been published in journals such as *Accounting Horizons*, *Decision Support Systems*, *IEEE Intelligent Systems*, *International Journal of Accounting Information Systems*, *Journal of Accountancy*, *Journal of Emerging Technologies in Accounting*, and *Journal of Information Systems*. He has also co-authored a research monograph published by the American Accounting Association titled *The REA Accounting Model as an Accounting and Economic Ontology*.

Guido has received numerous research, teaching, and service awards, including the 2012 American Accounting Association Strategic and Emerging Technologies Section Outstanding Contribution to Research Award, the 2015 University of Delaware's Excellence in Teaching Award, the 2018 American Accounting Association Outstanding Service Award, and the 2022 American Accounting Association/J. Michael and Mary Anne Cook/Deloitte Foundation Prize for undergraduate teaching.

Guido is the former chair of the Technology Task Force for the Pathways Commission Recommendation 4 (Curriculum and Pedagogy) and currently serves as a Trustee on the AICPA Foundation Board.



© Margarita Maria Lenk

MARGARITA MARIA LENK'S signature sincere enthusiasm, caring ethic, integrity, inclusion values, and her intellectual curiosity have served the accounting profession and students for decades in her roles as a national and international accounting professional, instructor, author, mentor, coach, consultant, presenter, and researcher. Margarita studied accounting, computer science, psychology, and economics, with degrees and teaching awards from the University of Central Florida, the University of North Carolina at Chapel Hill, and the University of South Carolina. She is a Colorado State University faculty emeritus.

Margarita integrates ethics, critical thinking, risk management, and data-driven decision-making perspectives in all her work. Her expertise includes data analytics, artificial intelligence, AIS security, governance and internal controls, sustainability accounting, social media risk management, service-learning, and university-community engagement and research partnerships. Her native Argentine innovative nature, strong service ethic and belief in our potential and responsibility to improve the world have always informed her accounting perspective and instructional philosophies towards developing future leaders.

Margarita has published over 50 interdisciplinary research articles in journals such as *Journal of Information Systems*, *Strategic Finance*, *Review of Accounting Information Systems*, *Journal of Production Planning and Control*, *Issues in Accounting Education*, *Journal of Information Systems Education*, *The Michigan Journal of Community Service Learning*, and *Business Communication Quarterly*. She has won over 30 instruction and mentoring related awards, including the 2019 Michael J. and Mary Ann Cook/Deloitte Foundation/AAA Prize for Outstanding Accounting Undergraduate Professor, the 2009 Outstanding Accounting Professor in the state of Colorado, the CSU System Board of Governors Outstanding Undergraduate Professor, and the 2006 Best Information Systems Professor from the Strategic and Emerging Technologies of the American Accounting Association. In 2021, she was inducted into the PhD Project Hall of Fame for mentoring and coaching three decades of diverse accounting faculty and students.

From the Authors

Dear Students,

The accounting profession is in a transformative time with data and analytics at its center. The accountants of the future will be agile, creative, data literate professionals who use critical thinking and problem-solving to drive business and professional value. The materials presented within this course will help you become a successful accountant in this new data and analytic environment.

Our author team is passionate about your success, and we have each applied unique skill-sets and expertise to the resources created for this course. Our critical thinking framework and data analysis process model will help you develop the professional skills you need to plan data analysis projects, perform analyses, and report their results. Our pattern-based approach to analysis skills will help you effectively prepare data, calculate the information needed for analytical purposes, and explore data from a variety of perspectives. This approach has a hands-on emphasis that provides plenty of opportunities to learn new technology skills.

The course has several distinguishing features that will help you develop your critical thinking, data analysis, and communication skills:

- A critical thinking framework with six elements that can be applied to any data analysis process (SPARKS).
- A data analysis process model with three stages that reflects the process used in the accounting profession (Plan, Analyze, Report).
- A model for planning data analysis projects.
- A pattern-based approach to profiling, cleaning, and transforming data, calculating information for analytical purposes, and exploring key data relationships.
- Applications to five accounting areas: accounting information systems, auditing, financial, managerial, and tax accounting.
- Theories and tools to help you interpret the results of data analyses.
- Best practices for creating data visualizations and telling a compelling story with data.
- Exposure to emerging technologies and how they are changing accounting data analysis projects.

We believe that knowing how to use data to improve processes, enhance efficiencies and accuracy, assess and reduce risks, and evaluate issues and opportunities will maximize your professional and societal impacts and the effectiveness of your organizations. We also want you to develop the technical skills to accomplish these goals.

This course and its accompanying learning resources will help you:

- Develop a data analytics mindset.
- Follow sustainable, documented, and defensible data analysis processes.
- Integrate critical thinking into the data analysis process by understanding the stakeholders, purpose, analysis alternatives, risks, and knowledge needed to prepare, interpret, and communicate data analysis results.
- Develop foundational, professionally relevant technical skills you can apply across tools and accounting areas.
- Develop the technical agility and data acumen needed to create data analytics solutions for accounting problems.

We are excited you are using these resources and wish you tremendous success in your accounting career.

ANN DZURANIN, PhD, CPA
GUIDO GEERTS, PhD
MARGARITA LENK, PhD, CMA

Pedagogical Framework

This course uses a process-based approach to data analysis by systematically introducing, explaining, and applying key concepts at each stage of the data analysis process.

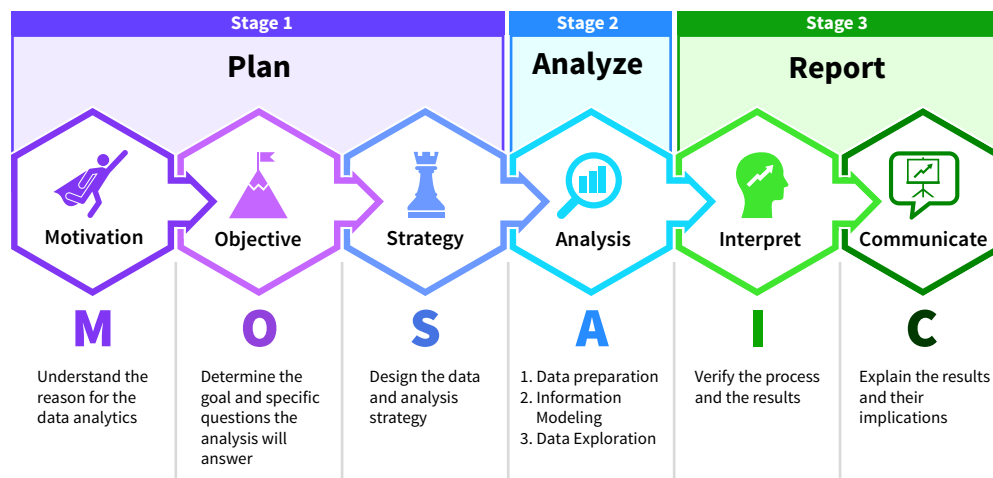
Laying the Foundation

Chapter 1 provides an overview of how data and analytics are being used in the accounting profession, the data analysis process, and the benefits of integrating critical thinking into that process. The visual of the Chapter Roadmap feature summarizes the topics covered in Chapter 2, which is a review of foundational data analysis skills to prepare the student for the rest of the course.

Chapter Roadmap		
LEARNING OBJECTIVES	TOPICS	APPLY IT
LO 2.1 Describe how data is stored in and extracted from relational databases.	- Relational Databases - Joining Tables	Identify Primary and Foreign Keys (Example: Accounting Information Systems)
LO 2.2 Explain how functions help answer data analysis questions.	- Basic Functions for Data Analysis - Applying Excel Basic Functions	Analyze Sales Transactions with Excel Functions (Example: Financial and Managerial Accounting)
LO 2.3 Illustrate how pivot tables organize and filter data.	- Using Pivot Tables - Filtering Pivot Tables	Analyze Sales with Excel PivotTables (Example: Financial and Managerial Accounting)
LO 2.4 Identify descriptive measures used to perform data analysis.	- Measures of Location - Measures of Dispersion - Measures of Shape - Correlation Analysis	Use Descriptive Statistics to Audit Warranty Expense (Example: Auditing)
LO 2.5 Summarize how data visualization explores and explains data.	- Making Sense of Large Data Sets - Visualizations and When to Use Them - Microsoft Excel Visualizations	Analyze Product Costs with Data Visualization (Example: Managerial Accounting)

Integrating Critical Thinking into the Data Analysis Process

Chapters 3 through 9 focus on the three stages of the data analysis process—Plan, Analyze, and Report.



- Chapter 3 develops the first two steps of the planning stage by explaining how motivation, or the “why” behind the project, influences the project’s objective. This objective is the foundation for developing the specific analysis questions that must be answered. How to design data and analysis strategies to answer these questions and achieve the project’s objective is covered in Chapter 4.
- Chapters 5 through 7 describe the analysis stage, demonstrating how patterns can be used to prepare data for analysis, build information models, and explore data.
- Chapters 8 and 9 detail how to report analyses results by accurately interpreting and effectively communicating them.

Finally, Chapter 10 discusses how changing data characteristics and recently developed technologies and tools are impacting how accounting professionals are performing data analysis.



Six essential elements of critical thinking are integrated into each chapter. By consistently applying the SPARKS critical thinking framework to data analysis topics, students refine their ability to think critically as they develop their data analysis skills.

Using a Pattern-Based Approach

The three analysis chapters use a unique approach to learning data analytics—patterns, which the student can leverage across data analytics projects. Each pattern represents a real-world scenario commonly encountered during data analysis and presents best practices and techniques to address such a scenario.

Patterns are introduced, explained, and summarized at key points, including at the conclusion of discussions and as part of the chapter summaries. This diverse set of patterns provide the foundational knowledge needed to be a successful data analyst.

ILLUSTRATION 7.46 Overview of Data Exploration Patterns

Id	Data Relationship	Description	Visualization Examples (Bold indicates chapter example)
Foundational Data Relationships			
1	Nominal Comparison	Compares the values of a nominal variable based on the values of a second, numeric variable.	Bar chart, column chart , dot plot, lollipop chart
2	Distribution	Shows how the values of a numeric variable are distributed, or spread.	Histogram, box-and-whisker (boxplot) chart , violin plot
3	Deviation	Shows how a set of actual values deviate from their reference values such as budgeted or forecasted values.	Clustered bar and column charts , gauge, bullet chart
4	Ranking	Orders the values of a variable sequentially based on the values of a second variable.	Defined as part of a table, bar chart, or column chart
5	Part-To-Whole	Shows how each part compares to the whole and how the parts compare to one another.	Pie chart , donut chart, stacked bar chart, stacked column chart, treemap chart
6	Correlation	Indicates the degree to which two variables move in the same or the opposite direction.	Scatterplot
7	Time Series	Shows the values of a variable at sequential points in time.	Area chart, bar chart, column chart, line chart , sparkline chart, waterfall chart
8	Geospatial	Assigns numeric values to locations and encodes them through coloring (shade) and size (bubble size).	Maps
Integrated Data Relationships			
9	Composite Trends	Shows how a composite structure (part-to-whole) changes over time (time series).	Stacked column chart, 100% stacked column chart , stacked area chart
10	Pareto Analysis	Illustrates the importance of different categories (nominal comparison), ranks them (ranking), and shows their cumulative percentage (part-to-whole).	Pareto chart, line and column chart

Pattern 1 Summary	
Problem	All the data did not transfer.
Detect (Data Profiling)	Compare row counts.
Correct (ETL)	Add the missing rows.

Connecting to the Profession

Chapter features, examples, questions, and cases present a mix of accounting examples so students can see how topics can be applied in different accounting contexts.

Insights from Accounting Professionals

Each chapter opens with an insight from an accounting professional introducing key topics covered in the chapter. These engaging insights are a window into how data analysis is being used in the accounting profession.

Professional Insight: Do the Analysis Results Make Sense?

Megan, a senior tax accountant, works at one of the ten largest U.S. accounting firms.

When I first graduated, the firm I worked in outsourced all tax return and analysis preparation. Right away I was expected to review analyses prepared by other people and explain those results to the partner. It was really difficult to suddenly be in a review role rather than a preparer role. I knew how to check the numbers to see if they were correct, but interpreting what the results meant was much more challenging.

As I have transitioned into a senior role, I have seen the importance of critical thinking even more. My favorite staff to work with are those who review their results and ask questions. They don't just assume that, because the numbers were input correctly, the results are correct. **You can check all the numbers to see if they are correct, but when you do the analysis, you need to be able to determine if the results make sense.**

A staff member once input all the data correctly into the tax software and prepared a tax liability analysis that showed the client's tax liability went up. However, this didn't make sense because that client had a \$100 thousand loss from one of their businesses. So, the tax liability should have gone down. When I pointed this out, the staff member responded, "I was so caught up in the details that I rationalized the results instead of asking more questions." Stepping back to evaluate if something makes sense is difficult. As a student I didn't practice doing that. I was just focused on getting the right answers and getting a high grade.

Accounting Area Examples

Chapter topics are consistently applied to accounting areas such as financial, managerial, and tax accounting, as well as AIS and auditing. Application opportunities and end of chapter assessment are also tagged by accounting areas to help students practice applying data analysis to these respective accounting areas.

ILLUSTRATION 3.35 Motivations, Objectives, and Questions for Managerial Accounting Data Analytics

Motivation	Objective	Question
Identify and evaluate new market, product, service, and business process opportunities.	Determine whether to expand an existing product line.	What are the projected sales for the new product line?
Efficiently allocate resources for successful operations.	Evaluate the best use of materials for production.	Which products use the most materials?
Maximize revenue generation and cash flows.	Determine the most profitable product mix.	How many units of each product should we produce to optimize contribution margin?
Increase organizational efficiency by managing costs.	Determine potential cost cutting.	Which production line has the highest amount of waste?

BE 8.11 (LO 5) Tax Accounting One consideration of tax planning is understanding and projecting the amount of net revenue that a company expects for the upcoming year. Your team has created a what-if analysis to predict Super Scooters' net revenue for next year. Based on current year data, the team has performed three sensitivity analyses:

Apply It 1.2

Explain the Data Analysis Process for a Fraud Risk Assessment

Auditing LocalPics (LP) is a busy movie theater chain that does a lot of business on the weekends. LocalPics was tipped off by an employee that there was a cash shortage. The cash came from ticket sales. Unfortunately, the anonymous person who tipped them off was the person who was stealing the cash. The theaters the theft was happening. The CPA firm, CFW, had to investigate the theft. The firm found out that the person who was stealing the cash was the person who was tipped off. The firm then helped the company to recover the cash and prevent future thefts.

Identify the steps in the data analysis process for the explanation of each.

Pre-created homework assignments for specific areas of accounting are also available on Wiley's online learning platform.

Ch 8: Homework - Auditing	9	Homework	USE SET ▾
Ch 8: Homework - Financial Accounting	11	Homework	USE SET ▾
Ch 8: Homework - Managerial Accounting	7	Homework	USE SET ▾

Developing In-Demand Skills

Chapter features help students develop the skills employers are demanding from accounting professionals today.

Critical Thinking

In each chapter, Applying Critical Thinking feature boxes illustrate how a disciplined, reasoned approach to data analysis is the foundation of a data analytics mindset.

APPLYING CRITICAL THINKING 4.4: Threats to Data Strategy

Data risks can impact the accuracy and validity of analysis results. These risks must be evaluated so appropriate controls can be added to the project plan. Data strategy risks can include (Risks):

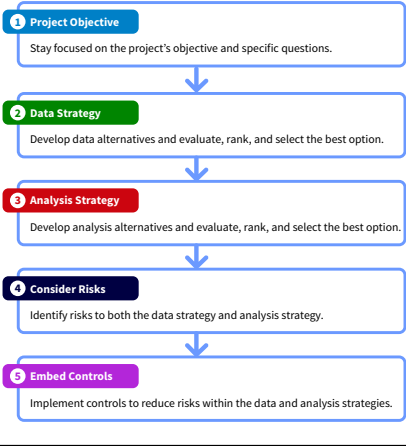
- Dirty data such as incomplete data, inaccurate data, and incorrectly formatted data fields.
- Data irrelevant to the objective of the project.
- Insufficient data for the analysis to be reliable.
- Data that is an unrepresentative sample of the underlying population.
- Errors in management estimates and assumptions about the data.

Chapters also demonstrate critical thinking application at key points and emphasize the benefits of evaluating a data analysis project from multiple angles at each process stage.

ILLUSTRATION 3.5 Thinking Critically About Motivations and Objectives

Critical Thinking Element	Why It Matters	Super Scooters Example
S Stakeholders	Considering the perspectives of stakeholders helps identify the objective and develop good questions.	The internal stakeholders are Lyla and Calvin. Their motivation is to better understand company performance. The objective is to describe current performance, diagnose issues related to poor performance, predict future sales, and make recommendations for improvements.
P Purpose	Critically thinking through the purpose of data analyses is the link between a defined objective and specific questions: • A purpose is the reason something is done or created. • An objective is something we plan to do, or a goal. The questions will be different depending on whether the purpose is to describe and diagnose current conditions or predict and prescribe future outcomes.	Based on the objectives, the purpose is to: • Describe what is happening now and diagnose issues related to poor performance. • Predict future sales. • Prescribe a profitable sales mix.
A Alternatives	Evaluating different methods of analysis increases the likelihood of selecting the best option for the project's purpose.	Develop several alternative methods for performing the analysis to describe performance over the past year. Consider how likely each option is to provide valuable information. Rank the alternatives and choose the one with the highest expected value.
R Risks	Being aware of potential risks makes it possible to also mitigate them when evaluating motivation, objectives, and questions.	It may be tempting to make Calvin and Lyla happy with a positive outlook in the predictive model. Be aware of that bias and develop objectives and questions that will not be influenced by it.
K Knowledge	Successfully completing a project requires confirming we have the necessary knowledge to do so or doing additional research to obtain that knowledge.	One purpose of the Super Scooters analysis is to predict future sales and expenses. If you do not know how to prepare a predictive analysis, you must do one of two things: • Hire someone to prepare the predictive analysis. • Learn how to do the analysis.
S Self-Reflection	Understanding the purpose of the analysis makes it easier to apply the experience of performing it to future data analytics contexts and tasks. Similarly, considering past experiences can help identify objectives and develop questions.	Assessing whether you should have performed the analyses may depend on whether the information lead to better decisions by management. If it did not, then consider why. Only then can you compare expected net benefits to the actual net benefits.

ILLUSTRATION 4.1 A Data Analysis Project Plan



Digital Skills

A tool-agnostic approach is the foundation for helping students develop the technical agility that accounting professionals working with data need. Chapters on Wiley’s online learning platform also include a video, created by the authors for students, which highlights how students can best use the resources in each chapter.



How to Walk-Throughs

How To Walk-Throughs appear at the end of each chapter. They are step-by-step guides for recreating chapter illustrations or examples using a technology tool covered in the course—Power BI, Alteryx, Tableau, or Excel. When possible, alternative technology options are also presented. (For example, a How To in Chapter 8 details how to create a frequency table using Power BI, but Excel and Tableau are also highlighted as alternatives). Walk-through videos to accompany the How To features are also available online.

Chapter How To

HOW TO 9.1 Create a Dashboard in Tableau

The dashboard in Illustration 9.13 was created using the visualization software Tableau. Follow these steps to create it yourself.

What You Need: **Data** The How To 9.1 data file.

STEP 1: Open the workbook to find the three visualizations that have already been created: profit margin by brand, gross sales by brand, and cost analysis.

STEP 2: Open a dashboard worksheet by selecting **Dashboard** in the main toolbar or by clicking the dashboard icon next to the worksheets (Illustration 9.49).

ILLUSTRATION 9.49 Tableau Dashboard Option in the Toolbar

STEP 3: In the dashboard worksheet, click and drag any of the visualizations listed under **Sheets** to the box **Drop Sheets Here**.

How To Video

How To 8.1: Create a Frequency Table Using Tableau

• Illustration 8.14: Super Scooters Model Frequency Distribution

Model	Frequency	Relative Frequency
Captain	1,010	27.7%
Celeritas	892	24.5%
Kicks	456	12.5%
Lazer	1,287	35.3%
Total Sales	3,645	100.0%

Apply It Features

A hands-on feature at the end of each learning objective, Apply Its offer regular opportunities for students to practice what they have learned. Videos detailing how to complete Apply Its are also available online.

Chapter Apply It

Apply It 8.5
Interpret Regression Results

Data **Managerial Accounting** DHI would like to understand what is driving total expenses for the hotel chain. Luciana feels strongly that the following variables have the most influence on expenses:

- Age of the hotel
- The number of maintenance employees
- Total housekeeping hours
- Total rooms rented

She has asked you to prepare a regression model using those variables to predict expenses. The following illustration is the result of that model.

Housekeeping Hours and Rooms Rented by Location Regression Model

Regression Statistics				
Multiple R	0.7215151			
R Square	0.520584			
Adjusted R Square	0.4714132			
Standard Error	93901.466			
Observations	44			

ANOVA					
	df	SS	MS	F	Significance F
Regression	4	3.73412E+11	9.3353E+10	10.58724498	6.62556E-06
Residual	39	3.43882E+11	8817485253		
Total	43	7.17293E+11			

	Coefficients	Standard Error	t Stat	P-value
Intercept	413314.47	107148.5543	3.85739665	0.000418405
Room Rentals	11.015461	2.918679199	3.77412543	0.000534577
Hours Worked, Housekeeping	6.0934507	5.531393108	1.1016123	0.277382666
Age	-3565.695	1674.938087	-2.12885155	0.039637083
Employees, Maintenance	16762.392	6906.902129	2.42690446	0.019949355

- What does the adjusted R square reveal about the model?
- Is the model better than not having a model?
- Are there any variables you would recommend Luciana remove from the model? Why?

SOLUTION

- The adjusted R square is 0.471, meaning that room rentals, housekeeping hours worked, the age of the hotel, and the number of maintenance employees can explain 47.1 % of total expenses.
- The model is significant – Significance F is less than 0.05 – so, it is better than no model at all.
- The p-value for Hours Worked, Housekeeping is greater than 0.05, so it should be removed from the model.

Apply It Video

Apply It 8.2
Evaluate a Cost Analysis

Auditing Denton Hospitality, Inc. (DHI) CEO Luciana Garcia is concerned about growing costs in some of the hotels and has asked the internal audit department to evaluate which hotels have increasing costs over the past two years. As an internal auditor for DHI, your manager asked you to prepare and interpret an analysis of costs by hotel for the past two years and identify the locations with the highest increases.

List each critical thinking element and identify how you would apply it to this example.

Robust Data Sets

Becoming technically agile requires working with data, so we offer over 125 unique data sets that include examples from small businesses, manufacturing, hotels, restaurants, services, hospitals, municipalities, state taxes, retail, financial services, and wholesale companies. Data tags attached to chapter examples and end of chapter material indicate when students can access related data on Wiley's online learning platform.

Data Let's use a case to illustrate how to apply data preparation patterns in a real-world scenario. You can also use the available data to work through each pattern yourself.

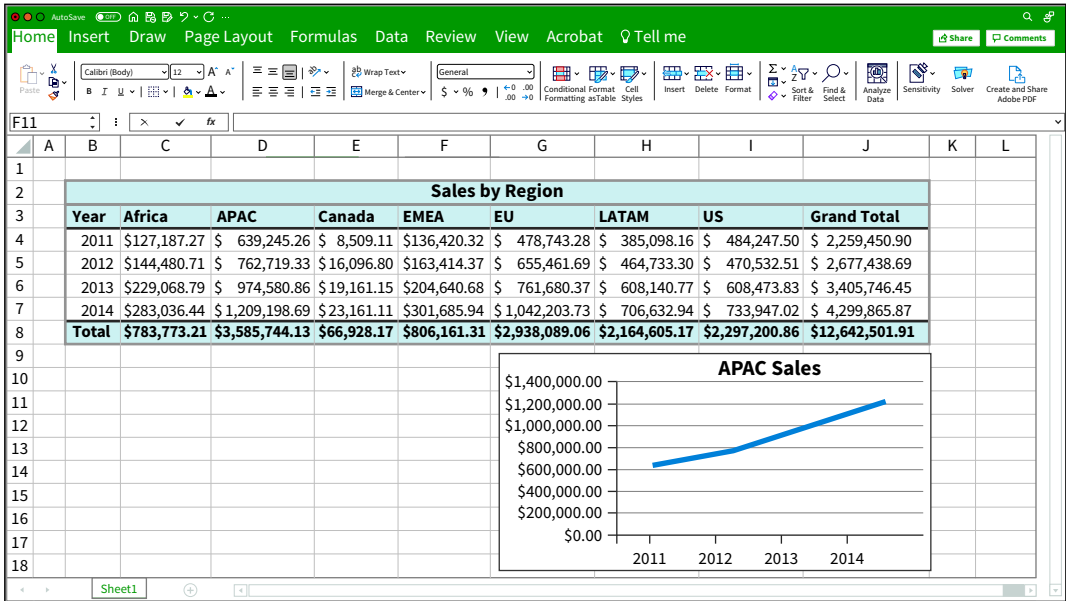
EX 9.6 (LO 5) Data **Financial Accounting** Create Interactive Visualizations U.S. Outdoor Adventures would like to use an interactive dashboard to evaluate product sales and profits. Review the static visualizations in one of the Excel, PowerBI, or Tableau files prepared by the accounting department at U.S. Outdoor Adventures. Convert the static visualizations into interactive visualizations that management can use to monitor the profitability of products by sub-category and location.

Data Analysis Technology Bootcamps

As an additional resource, a series of bootcamps are available on Wiley’s online platform to help students independently develop the skills to use data analytics software tools. Bootcamps include modules for additional Excel, Tableau, Power BI, and Alteryx.

Each module contains:

- Video walk-throughs demonstrating a specific task.
- Multiple choice questions to assess comprehension.
- Application exercises to apply what has been learned.
- Solution walk-through videos to check understanding.



Data Literacy and Communication Skills

The ability to work and communicate with data is one of the top skills demanded by employers. Learning to communicate with data can be challenging. We demystify creating effective data visualizations by providing easy-to-understand decision trees, best practice summaries, and data story guide-lines. After completing the course, students will be able to identify misleading visualizations and tell an accurate and engaging data story.

ILLUSTRATION 2.58 Guide to Visualizations that Show Composition, Relationships, and Distributions

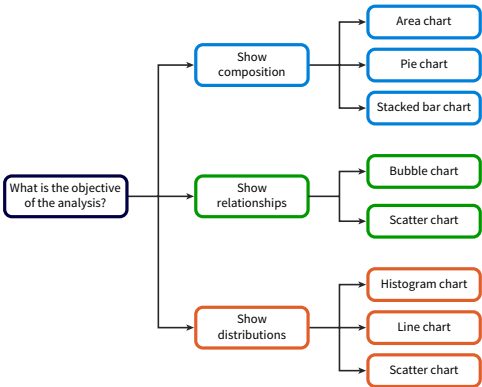


ILLUSTRATION 9.40 Best Practices by Visualization Type

Visualization	Best Practices
Area Chart	• Do not use with more than four categories to avoid confusion and clutter. • Start the y-axis at zero. • Put highly variable data on the top and data with low variability on the bottom.
Bar Chart	• Use horizontal bars if there are more than 7 categories or long category labels. • Use horizontal labels for better readability. • Space bars appropriately and consistently. • Use color sparingly, or as an accent. • Always have a zero baseline (the y-axis begins at zero). • Compare 2–7 categories with vertical columns.
Column Chart	

Each chapter helps students develop data literacy by giving them both opportunities to work with data and offering guidelines for doing so.

ILLUSTRATION 4.14 Analyses Strategies for Descriptive and Diagnostic Analytics

Potential Analyses	Data Measurement Scale			
	Categorical	Ordinal	Interval	Ratio
Central Tendency				
Mode	✓	✓	✓	✓
Median	✗	✗	✓	✓
Mean	✗	✗	✓	✓
Frequency Distributions				
Both counts and %s:				
Group frequency	✓	✓	✓	✓
Cumulative frequency	✓	✓	✓	✓
Relative frequency	✓	✓	✓	✓

(Continued)

End of Chapter Assessment

In addition to a chapter summary organized by learning objective and a list of key terms, the end of chapter material offers a variety of assessment opportunities.

Questions, Exercises, and Cases

A number of questions are available at the end of each chapter, including multiple choice, review questions, brief exercises, exercises, problems, and professional application cases to help students practice and assess their understanding of data concepts and application. “File upload” question types, available online, allow students to deliver and comment on the data analysis software files they create using the data sets associated with the questions.

[View Policies](#)

Current Attempt in Progress

Use the U.S. Outdoor Adventure datafile and recreate the scatterplot and descriptive statistics. Upload an excel or zip file that contains your s and descriptive statistics.

[DA_CH08_EOC_EX 8.15 - US Outdoor Adventure Stores.xlsx](#)

Supported file formats: Excel, ZIP

Press **T** to enter descriptions for the uploaded files (250 characters max).

The maximum number of files have been selected.

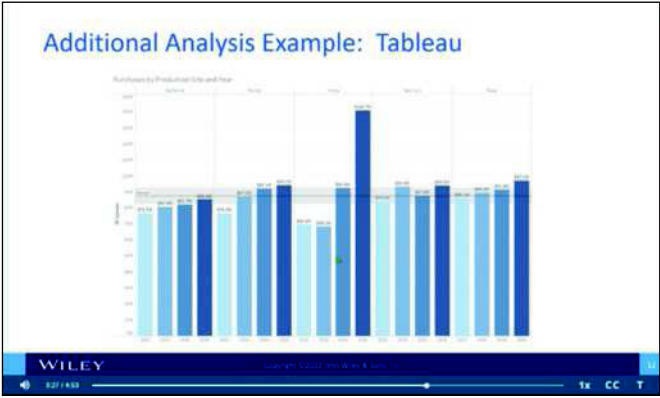
-  Data Analysis.xlsx

This is a scatterplot in Excel. Note how sales are decreasing.
-  Data Analysis.zip

This is the same scatterplot in Tableau. Note how sales are decreasing.

2 / 2 File Limit

Instructors can elect to make solution walk-through videos for selected end of chapter questions available to students online.



PR 8.1 (LO 1) Data Financial Accounting Interpret Descriptive Analytics Management at U.S. Outdoor Adventures is evaluating their shipping policy to customers. Currently, the company offers free shipping, but they are considering charging customers for shipping. Management would like to better understand how shipping costs are changing and if there is a relationship between the type of shipping method and order priority.

Products are shipped either first class, same day, second class, or standard class. The method of shipping is decided by the customer but is usually related to the order priority. The priority categories are critical, low, high or medium.

The data analysis team provided the following descriptive analyses, which indicate that total shipping costs increased from 2022 to 2025. They also prepared a frequency distribution of the number of shipments by priority and shipping mode for 2022, 2023, 2024, and 2025 combined.

Year	Total Shipping Cost
2022	\$ 49,769.30
2023	\$ 50,198.57
2024	\$ 60,925.19
2025	\$ 77,280.73
Grand Total	\$ 238,173.79

Priority and Shipping Mode	Number of Shipments
Critical	783
First Class	340
Same Day	154
Second Class	289
High	3,069
First Class	703
Same Day	273
Second Class	744
Standard Class	1,349
Low	432
Standard Class	432
Medium	5,710
First Class	495
Same Day	116
Second Class	912
Standard Class	4,187
Grand Total	9,994

1. Identify the following:

- Who are the stakeholders ?
- What was the purpose for analysis?
- Are there alternative explanations or analyses?
- Are there any risks that should be
- What knowledge did the preparer need and what knowledge does the interpreter need to understand the analysis?
- How can you use your prior experience or apply this experience to future analyses through self-reflection?

Professional Application Case

Every chapter concludes with a Professional Application Case. The case provides data for one company that can be used to perform analyses in each area of accounting. By focusing on one company, students can see how data analysis can be applied across accounting areas within one organization.

Professional Application Case: Ortho Inc.



Data Ortho Inc. is a medical device manufacturer based in San Diego, California. The company produces and sells custom-made knee replacement implants:

- It is a privately held company and files U.S GAAP based financial statements with its bank as part of its debt covenants.
- The board of directors is very involved and continues to emphasize the importance of well-designed and effective internal controls.
- Ortho is a decentralized organization, with 5 manufacturing and sales sites across the United States employing a total of 28 product managers and 29 sales managers. Each site operates independently and is responsible for procurement of raw materials, production, and sales of finished products in its own geographic region. Sales managers maintain their own customer relationships, while product managers monitor inventory levels and order raw materials as needed.
- Silicone and plaster are used to create the molds for all the company's knee replacements. The materials poured into molds vary by product-some are metal, some are plastic, and some are ceramic.

The company has invested significantly in the development and maintenance of its information system to collect and report complete and accurate data. Company management uses the data captured in the information system to create reports and support decision-making. The data dictionary includes the most relevant data used by management, accountants, auditors, and other professionals.

Data Dictionary

MATERIAL MASTER TABLE

Name	Primary Key	Definition
MaterialCategory	☐	Constant stating raw material
Material	☒	Material item identifier
MaterialDescription	☐	Description of material

Notes: The Material Master table contains information about different raw materials purchased by Ortho.

Le Grind Continuing Case

Available online, this running case tells the story of Le Grind, a regional commercial distributor of organic, fair trade certified coffee beans in the United States. Students can work

through the three stages of the data analysis process (Plan, Analyze, and Report). By the end of Chapter 9, students will have completed a data analysis of gross margin and prepared a communication of the results.

Le Grind Continuing Case: Verify and Interpret the Gross Profit Analysis

Data Visit Wiley's online learning platform for additional questions, data, and more details about the continuing case.



For Instructors

Providing useful instructional support is just as important as creating quality student resources. To further support instructors, the following resources have been created:

- **Instructor Guide:** Curated by the authors to include suggestions for how to best teach the content using the provided chapter-specific resources.
- **Chapter Lesson Plan:** Suggested instructional designs for each chapter.
- **Instructor Guide Videos:** Videos offer further guidance for implementing the chapter's resources.

End of Chapter Content

Question	Data	Course Application						
		General	Auth	Accounting	Financial	Managerial	Tax	Software
HE B-1		X						
HE B-2		X						
HE B-3					X			
HE B-4					X	X		
HE B-5		X						
HE B-6		X						
HE B-7				X				
HE B-8				X				
HE B-9					X			
HE B-10						X		
HE B-11							X	
EX B-1			X					
EX B-2			X	X				
EX B-3			X					
EX B-4					X			
EX B-5					X			
EX B-6		X			X	X		E
EX B-7			X	X				
EX B-8						X		
EX B-9	X				X			E
EX B-10	X					X		E
EX B-11						X		
EX B-12					X			
EX B-13	X				X			E
EX B-14					X			
EX B-15	X			X				E
EX B-16					X	X		
EX B-17	X					X		E
EX B-18	X					X		E
EX B-19	X				X			E
EX B-20					X	X		
PM B-1	X				X			E
PM B-2	X				X	X		E
PM - Audit	X			X				E, P, T
PM - Financial	X				X			E, P, T

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- **Solutions Manual:** Offers solutions to the end of chapter material for each chapter.
- **Test Bank:** Each chapter has a variety of questions with accompanying data sets available for assessment. Question types include multiple choice, multiple select, short answer, brief exercises, and exercises.
- **PowerPoint:** Lecture slides provide an overview of chapter content.

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We would not have been able to publish *Data and Analytics in Accounting* without the tremendous contributions of our advisory board members, reviewers, accuracy checkers, class testers, and the students who provided feedback during the creation of this product. We appreciate the recommendations and constructive comments from everyone. We also thank the instructors who participated in development and authoring activities for this edition. Diane Janvrin, Jennifer Riley, and Juergen Sidgman read and commented on every chapter. Matt Pickard and Vincent Shea provided helpful feedback on the continuing case. Special thanks goes to Renee Olvera for her valuable contributions to the end of chapter content and her thoughtful review of our online material.

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